CÀI ĐẶT APACHE HIVE

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1. Cài đặt Java

- Đã thực hiện trong các bài tập trước

```
hadoopminhchau@minhchau-master:~$ java -version
openjdk version "1.8.0_422"
OpenJDK Runtime Environment (build 1.8.0_422-8u422-b05-1~24.04-b05)
OpenJDK 64-Bit Server VM (build 25.422-b05, mixed mode)
hadoopminhchau@minhchau-master:~$
```

2. Cài đặt Hadoop

- Đã thực hiện trong các bài tập trước

```
hadoopminhchau@minhchau-master:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoopminhchau in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [minhchau-master]
Starting datanodes
Starting secondary namenodes [minhchau-master]
Starting resourcemanager
Starting nodemanagers
hadoopminhchau@minhchau-master:~$ jps
2196 ResourceManager
2020 SecondaryNameNode
2495 Jps
1791 NameNode
hadoopminhchau@minhchau-master:~$
```

```
hadoopminhchau@minhchau-slave1:~$ jps
2066 DataNode
2210 NodeManager
2328 Jps
hadoopminhchau@minhchau-slave1:~$ _
```

3. Download và cài đặt Apache Hive

Download Apache Hive 4.0.0

```
$ wget https://dlcdn.apache.org/hive/hive-4.0.0/apache-hive-4.0.0-bin.tar.gz
```

Giải nén và đổi tên thư mục

```
$ tar -xzf apache-hive-4.0.0-bin.tar.gz
$ mv apache-hive-4.0.0-bin hive
```

```
hadoopminhchau@minhchau-master:~$ ls
apache-hive-4.0.0-bin.tar.gz  hadoop  hadoop-3.4.0.tar.gz  hive  id.pig  passwd  pig  pig-0.17.0.tar.gz  sample.txt  tmp
hadoopminhchau@minhchau-master:~$
```

Cài đặt biến môi trường cho Hive

```
$ vim ~/.bashrc
```

```
export JAVA_HOME=/usr/lib/jvm/java-1.8.0-openjdk-amd64
export HADOOP_HOME=/home/hadoopminhchau/hadoop
export PATH=$PATH:$HADOOP_HOME/bin
export PATH=$PATH:$HADOOP_HOME/sbin
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
export HADOOP_YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"

export PIG_HOME=/home/hadoopminhchau/pig
export PATH=$PATH:$PIG_HOME/bin
export PIG_CLASSPATH=$HADOOP_CONF_DIR

export HIVE_HOME=/home/hadoopminhchau/hive
export PATH=$PATH:$HIVE_HOME/bin
export CLASSPATH=$CLASSPATH:/home/hadoopminhchau/hadoop/lib/*
export CLASSPATH=$CLASSPATH:/home/hadoopminhchau/hive/lib/*
~
-- INSERT --
```

4. Cấu hình Hive

- Metastore như là một kho trung tâm của Hive metadata, lưu trữ metadata cho các tables của Hive (như schema hay vị trí được lưu trữ).
- mv metastore_db metastore_db.tmp
- schematool -initSchema -dbType derby
- `hive --service hiveserver2 &`
- kill – 9 9832

Tạo và cấu hình hive-env.sh từ file mẫu

```
$ cd hive/conf
```

```
$ cp hive-env.sh.template hive-env.sh
```

```
hadoopminhchau@minhchau-master:~$ cd hive/conf/
hadoopminhchau@minhchau-master:~/hive/conf$ ls
beeline-log4j2.properties.template  hive-exec-log4j2.properties.template  llap-cli-log4j2.properties.template
hive-default.xml.template           hive-log4j2.properties.template       llap-daemon-log4j2.properties.template
hive-env.sh.template                ivysettings.xml                       parquet-logging.properties
hadoopminhchau@minhchau-master:~/hive/conf$ cp hive-env.sh.template hive-env.sh
hadoopminhchau@minhchau-master:~/hive/conf$ _
```

```
# Set HADOOP_HOME to point to a specific hadoop install directory
export HADOOP_HOME=/home/hadoopminhchau/hadoop

# Hive Configuration Directory can be controlled by:
# export HIVE_CONF_DIR=

# Folder containing extra libraries required for hive compilation/execution can be controlled by:
"hive-env.sh" 54L, 2378B written
hadoopminhchau@minhchau-master:~/hive/conf$ _
```

Tạo và cấu hình hive-site.xml từ file mẫu

```
$ cd hive/conf
$ cp hive-default.xml.template hive-site.xml
```

```
hadoopminhchau@minhchau-master:~/hive/conf$ ls
beeline-log4j2.properties.template    hive-log4j2.properties.template
hive-default.xml.template             ivysettings.xml
hive-env.sh                           llap-cli-log4j2.properties.template
hive-env.sh.template                  llap-daemon-log4j2.properties.template
hive-exec-log4j2.properties.template  parquet-logging.properties
hadoopminhchau@minhchau-master:~/hive/conf$ cp hive-default.xml.template hive-site.xml
hadoopminhchau@minhchau-master:~/hive/conf$ ls
beeline-log4j2.properties.template    hive-site.xml
hive-default.xml.template             ivysettings.xml
hive-env.sh                           llap-cli-log4j2.properties.template
hive-env.sh.template                  llap-daemon-log4j2.properties.template
hive-exec-log4j2.properties.template  parquet-logging.properties
hive-log4j2.properties.template
hadoopminhchau@minhchau-master:~/hive/conf$
```

Thêm vào nội dung sau:

```
  <property>
    <name>hive.exec.local.scratchdir</name>
    <value>/tmp/hive/hadoopminhchau</value>
    <description>Local scratch space for Hive jobs</description>
  </property>
  <property>
    <name>hive.downloaded.resources.dir</name>
    <value>/tmp/hive_io/${hive.session.id}_resources</value>
    <description>Temporary local directory for added resources in the remote file system.</description>
  </property>
```

```
  <property>
    <name>hive.querylog.location</name>
    <value>/tmp/hive/hadoopminhchau</value>
    <description>Location of Hive run time structured log file</description>
  </property>
```

```
<property>
  <name>hive.server2.logging.operation.log.location</name>
  <value>/tmp/hive/hadoopminhchau/operation_logs</value>
  <description>Top level directory where operation logs are stored if logging functionality is enabled</description>
</property>
```

Hive Database warehouse location

```
  <property>
    <name>hive.metastore.warehouse.dir</name>
    <value>/user/hive/warehouse</value>
    <description>location of default database for the warehouse</description>
  </property>
```

Hive Metastore database

```
  <property>
    <name>javax.jdo.option.ConnectionURL</name>
    <value>jdbc:derby:;databaseName=metastore_db;create=true</value>
    <description>
      JDBC connect string for a JDBC metastore.
      To use SSL to encrypt/authenticate the connection, provide database-specific SSL flag in the connection URL.
      For example, jdbc:postgresql://myhost/db?ssl=true for postgres database.
    </description>
  </property>
```

Tạo Hive Warehouse Directories

```
$ hdfs dfs -mkdir /user
$ hdfs dfs -mkdir /user/hive
$ hdfs dfs -mkdir /user/hive/warehouse
$ hdfs dfs -mkdir /user/tmp
$ hdfs dfs -chmod g+w /user/hive/warehouse
$ hdfs dfs -chmod g+w /user/tmp
```

```
hadoopminhchau@minhchau-master:~$ hdfs dfs -mkdir /user
hadoopminhchau@minhchau-master:~$ hdfs dfs -mkdir /user/hive
hadoopminhchau@minhchau-master:~$ hdfs dfs -mkdir /user/hive/warehouse
hadoopminhchau@minhchau-master:~$ hdfs dfs -mkdir /user/tmp
hadoopminhchau@minhchau-master:~$ hdfs dfs -chmod g+w /user/hive/warehouse
hadoopminhchau@minhchau-master:~$ hdfs dfs -chmod g+w /user/tmp
hadoopminhchau@minhchau-master:~$
```

Tạo Hive Metastore Derby Database

schematool -initSchema -dbType derby

```
hadoopminhchau@minhchau-master:~$ schematool -initSchema -dbType derby
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hadoopminhchau/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoopminhchau/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Initializing the schema to: 4.0.0
Metastore connection URL:        jdbc:derby:;databaseName=metastore_db;create=true
Metastore connection Driver :    org.apache.derby.jdbc.EmbeddedDriver
Metastore connection User:       APP
Starting metastore schema initialization to 4.0.0
Initialization script hive-schema-4.0.0.derby.sql
```

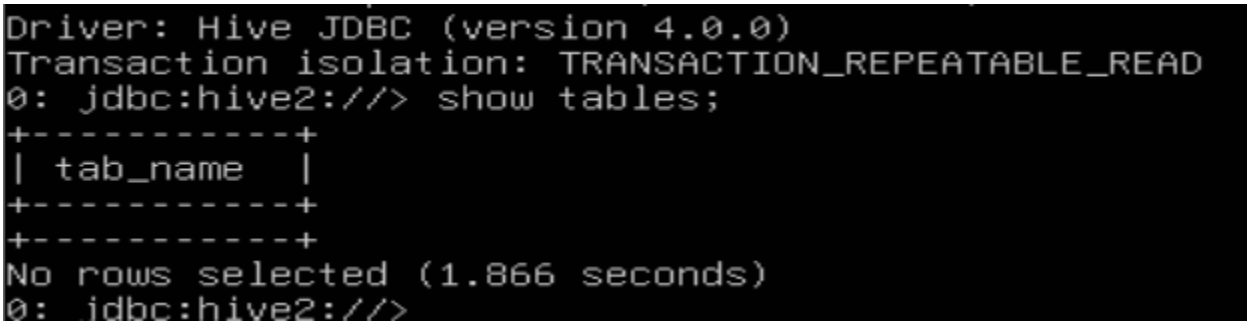
```
Initialization script completed
hadoopminhchau@minhchau-master:~$
```

Using Beeline in Embedded Mode

Khi kết nối vào HiveServer2 sẽ yêu cầu nhập user và pass. Có thể dùng user hadoopminhchau(pass là hadoopminhchau) hoặc user mặc định Hive cung cấp là scott (pass là tiger) để vào.

```
Initialization script completed
hadoopminhchau@minhchau-master:~$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hadoopminhchau/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoopminhchau/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hadoopminhchau/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoopminhchau/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Beeline version 4.0.0 by Apache Hive
beeline>
```





Khi không sử dụng thì dùng lệnh !close để ngắt kết nối đến HiveServer2 và !quit để thoát Hive.

